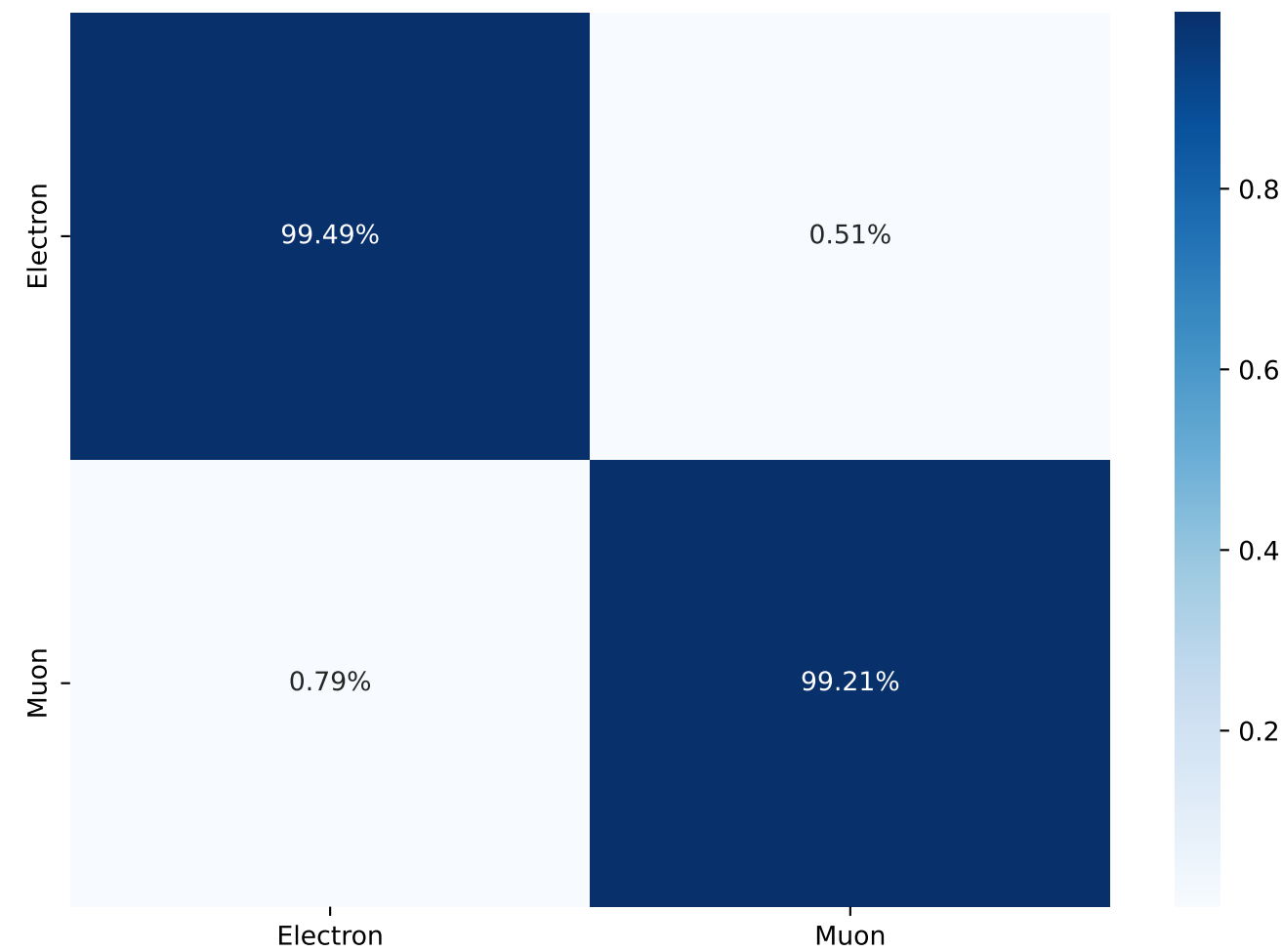
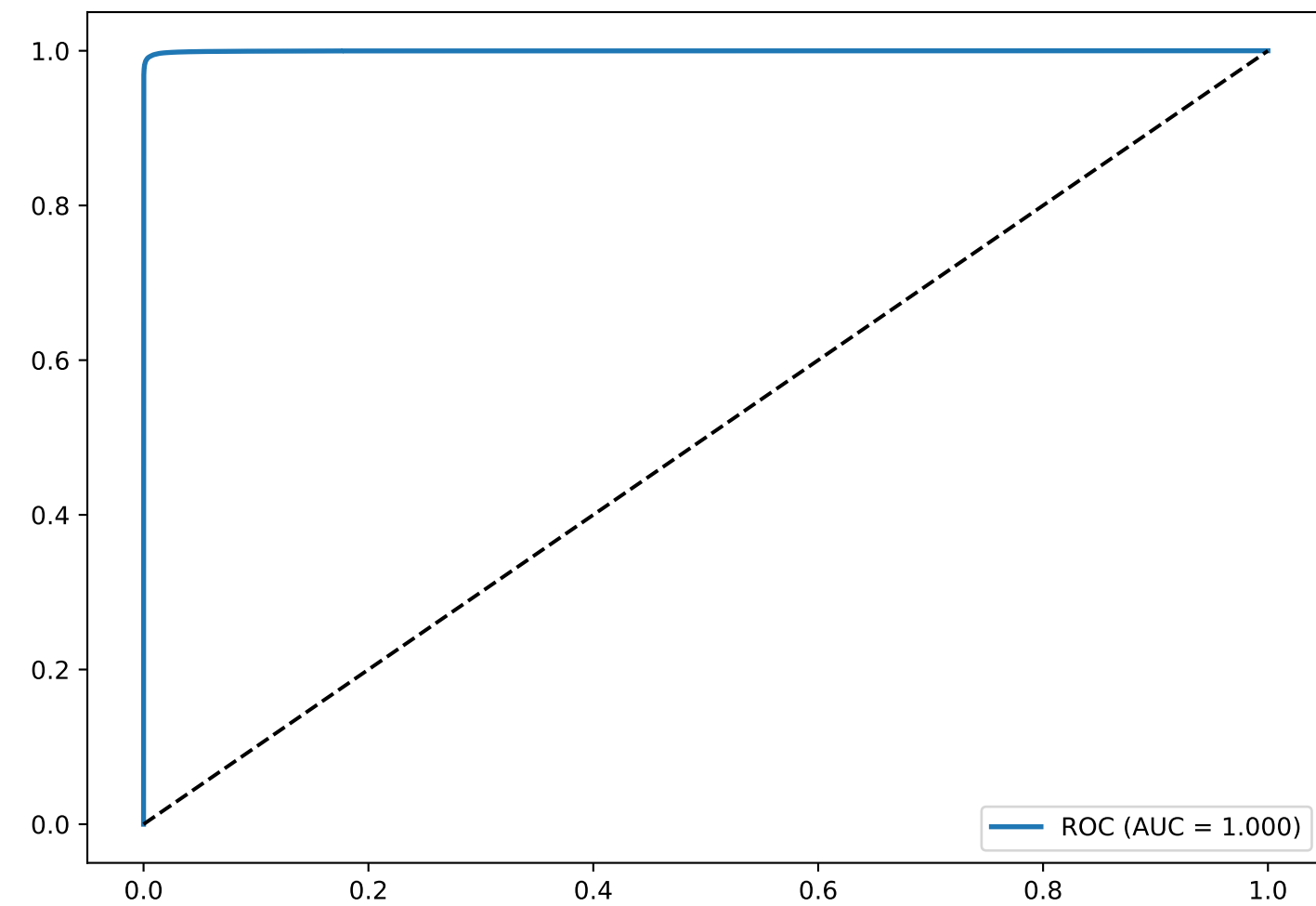


# EtaRange = FullRange (TEST DATA) - Accuracy: 0.993

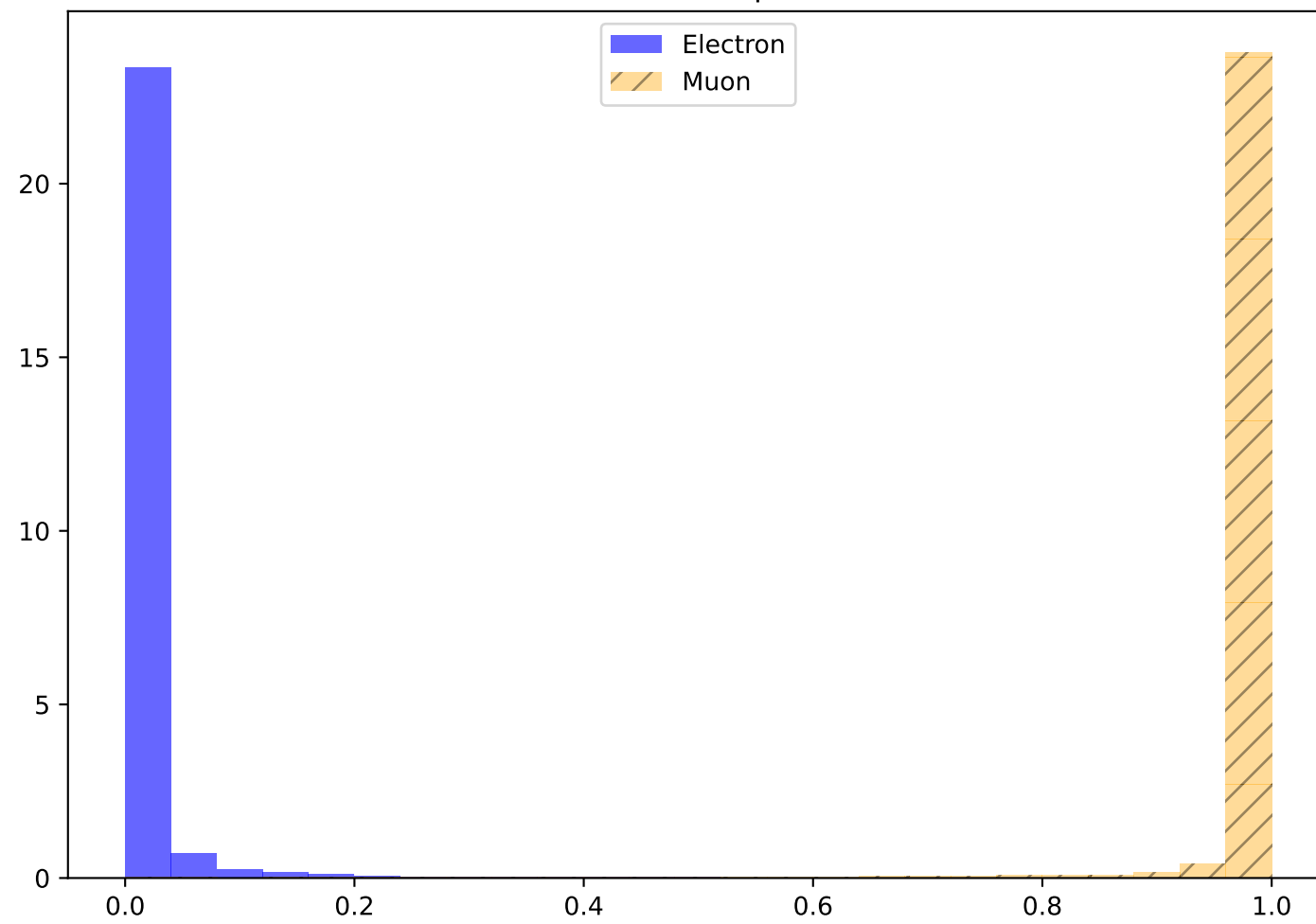
Confusion Matrix



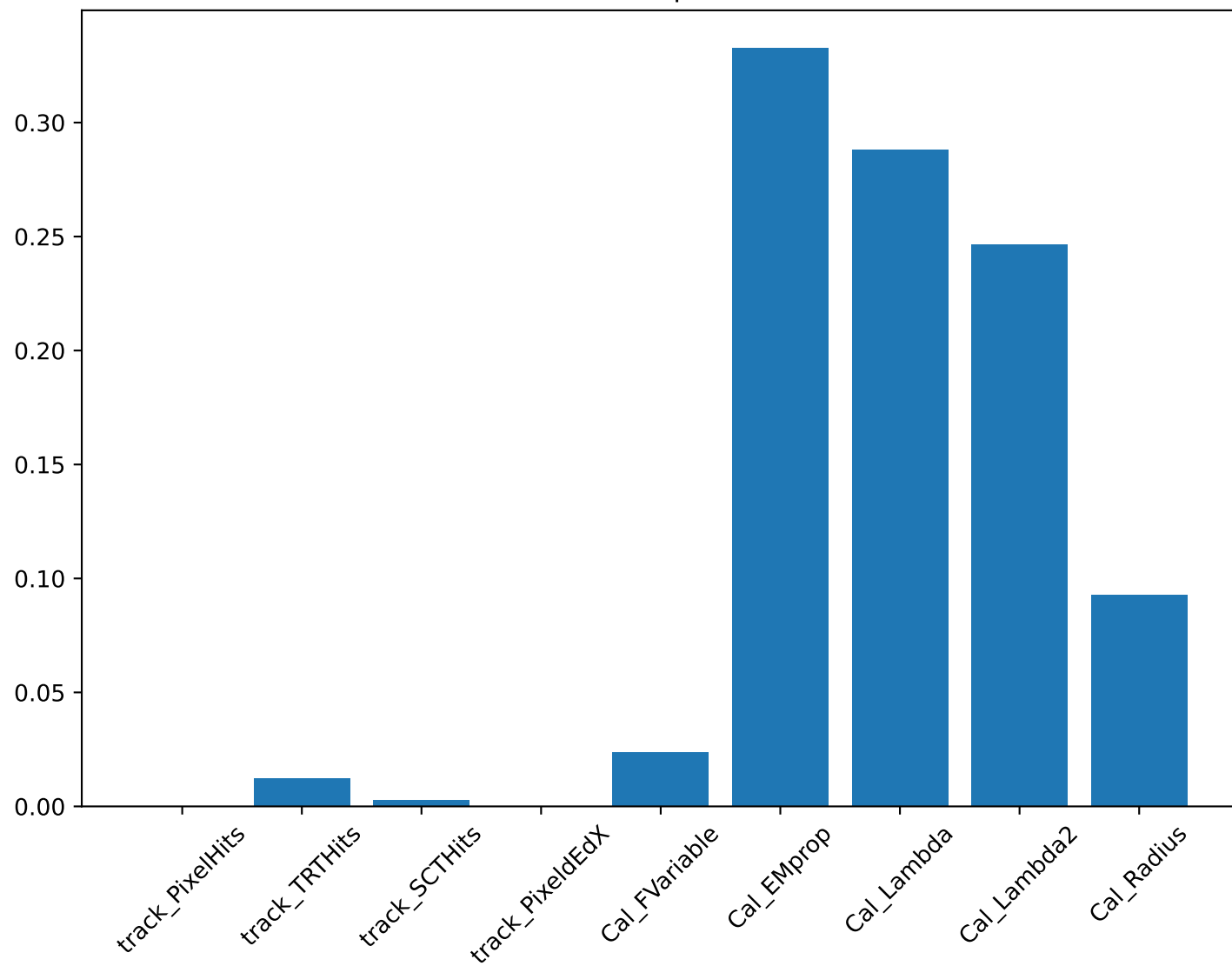
ROC Curve



Classifier Response

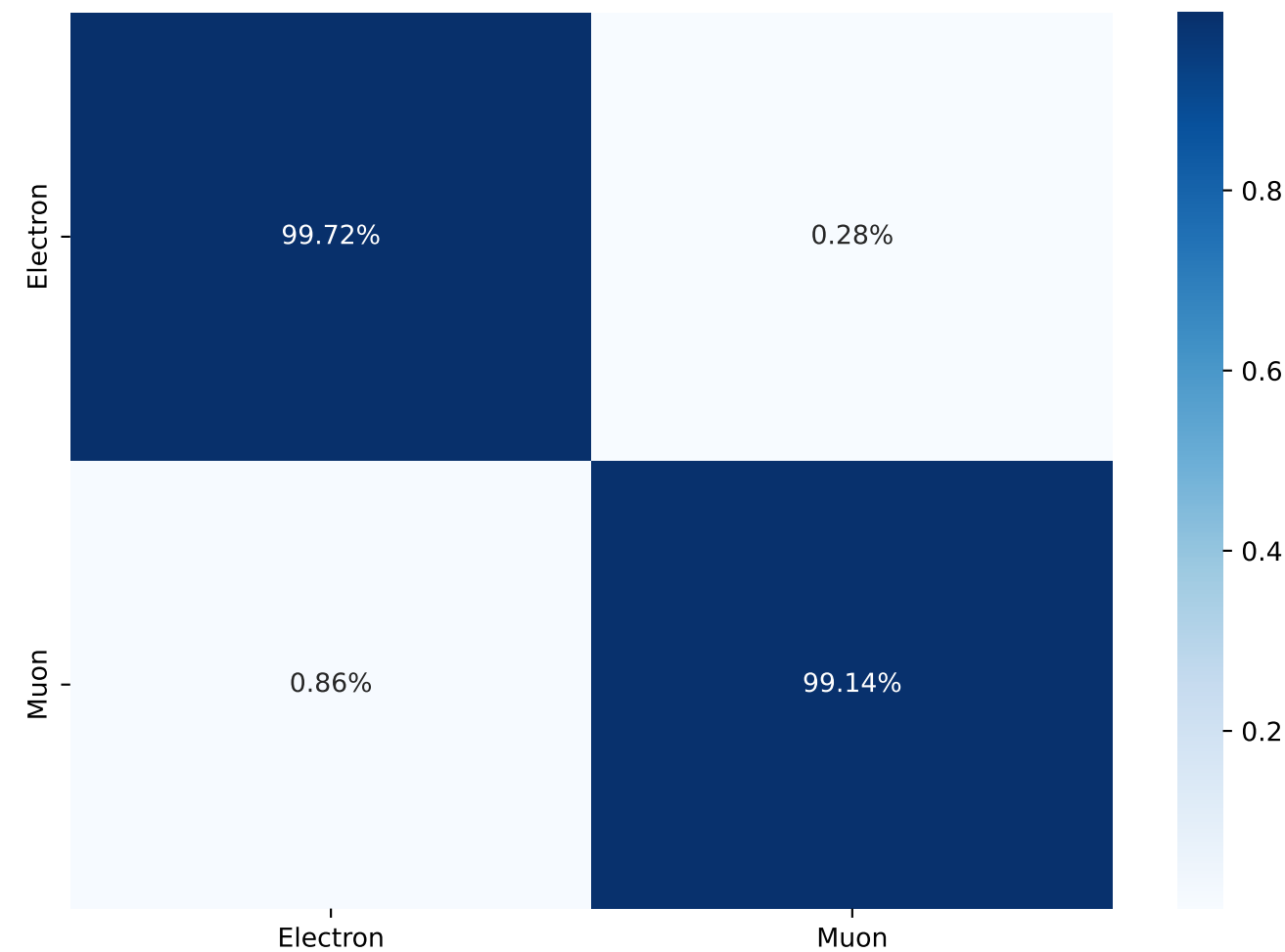


Feature Importance

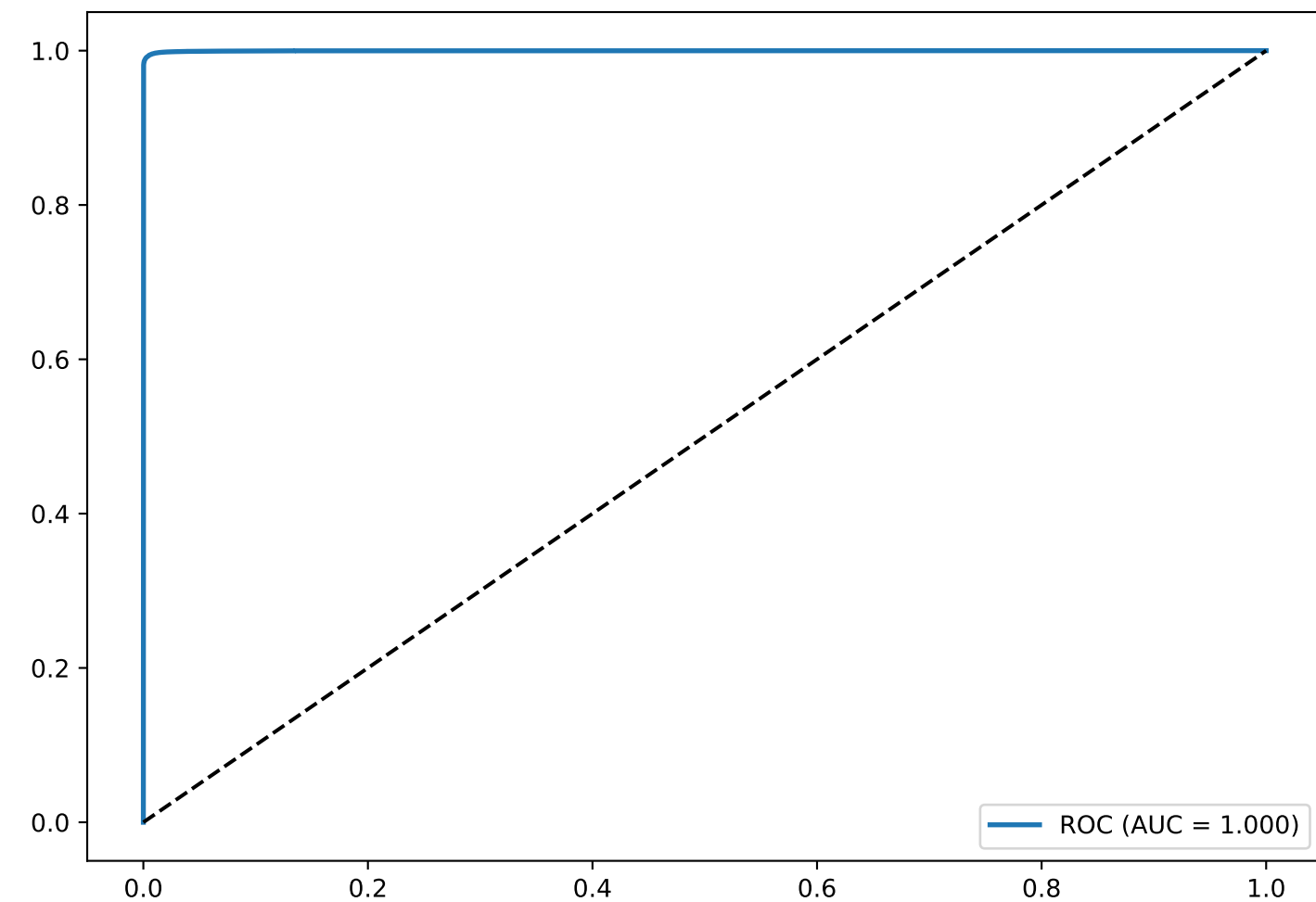


# EtaRange = <1.0 (TEST DATA) - Accuracy: 0.993

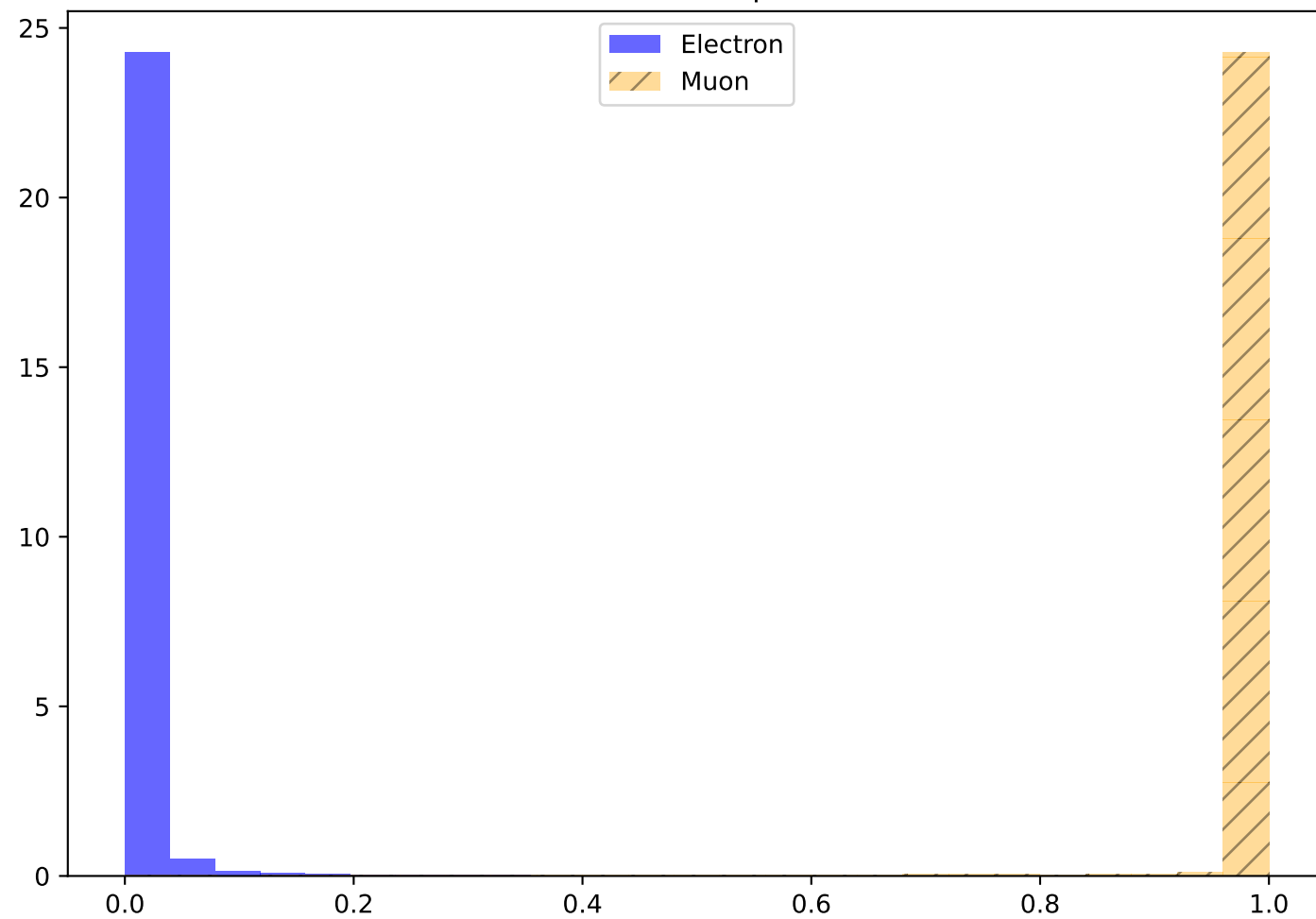
Confusion Matrix



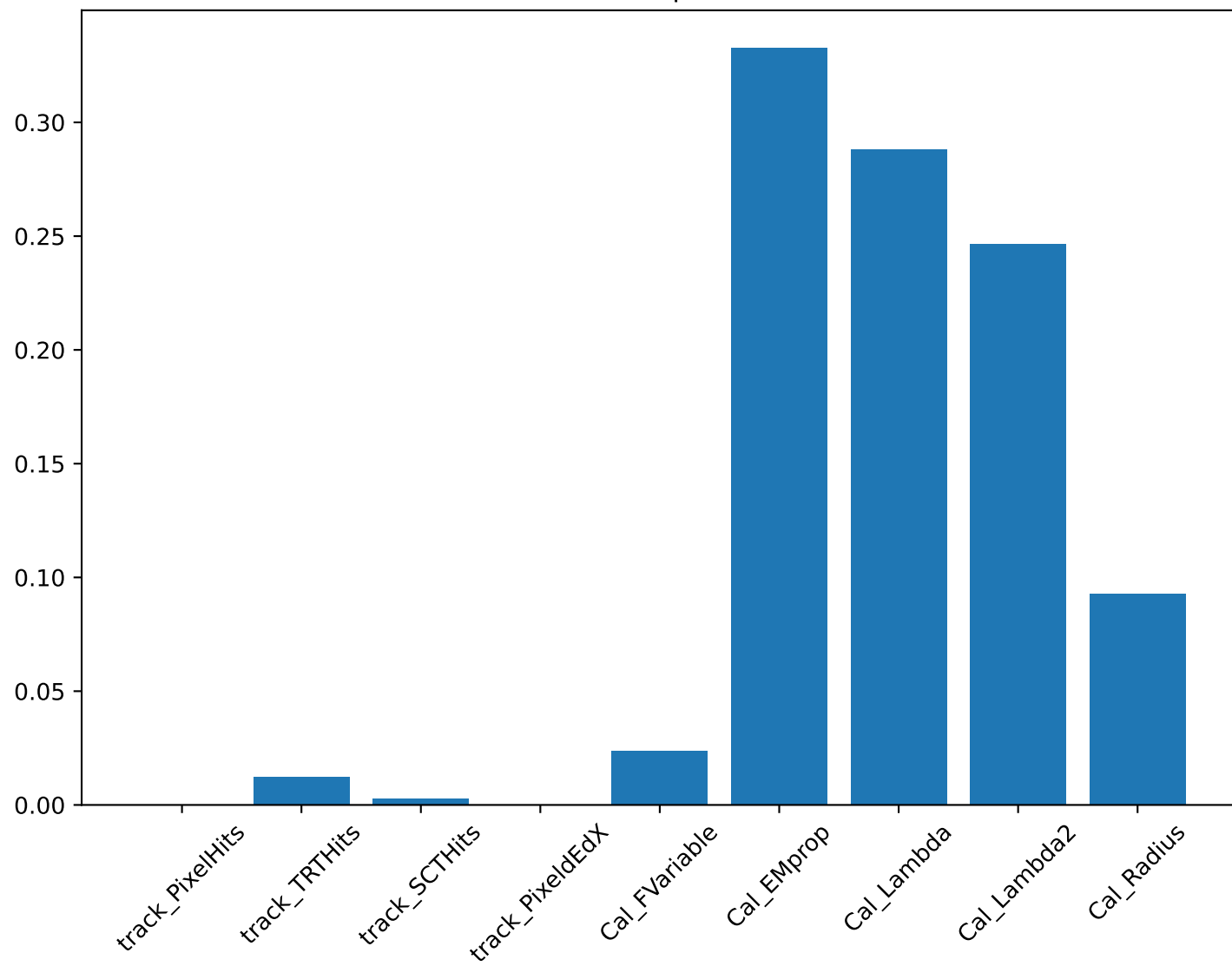
ROC Curve



Classifier Response

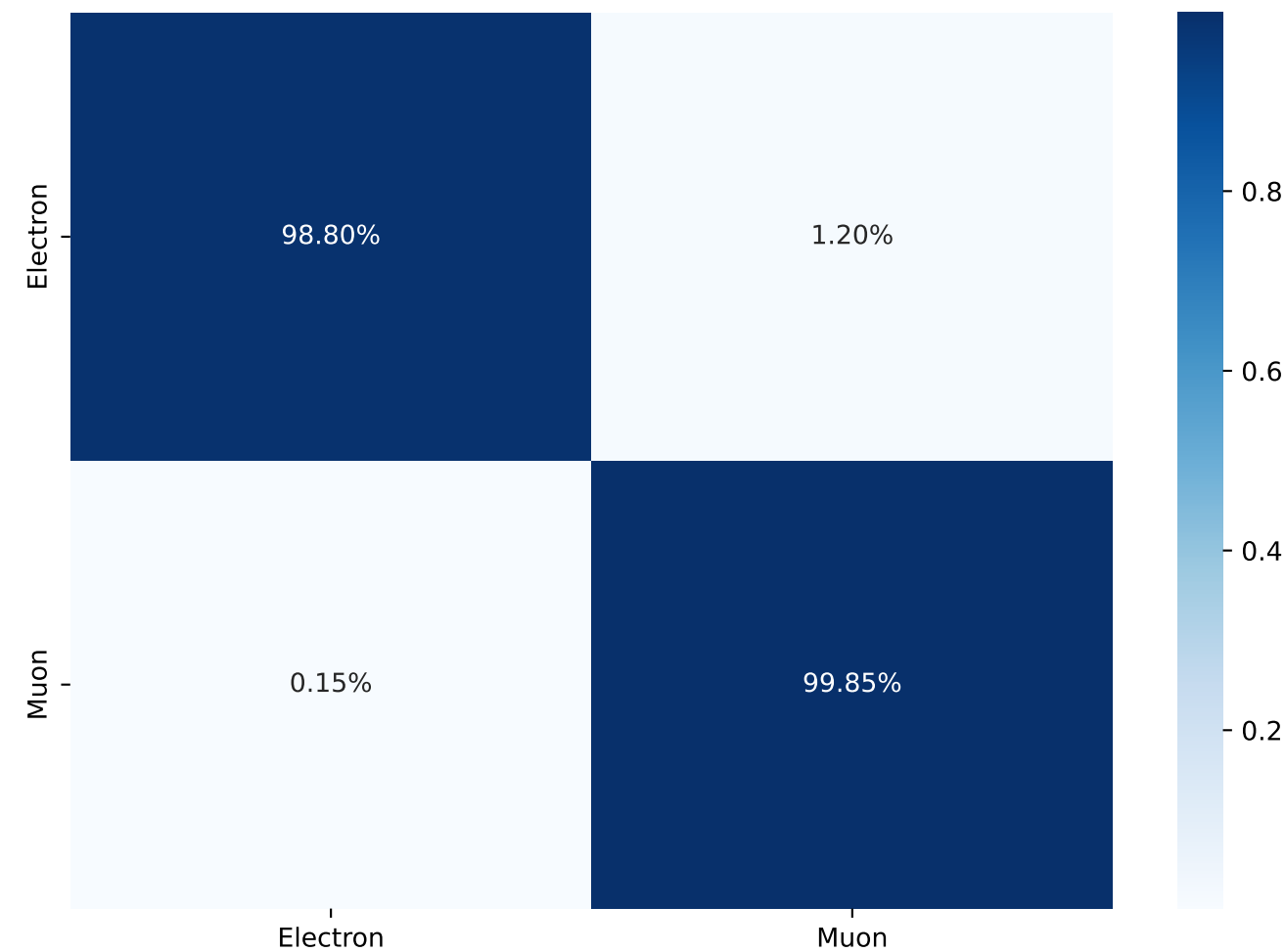


Feature Importance

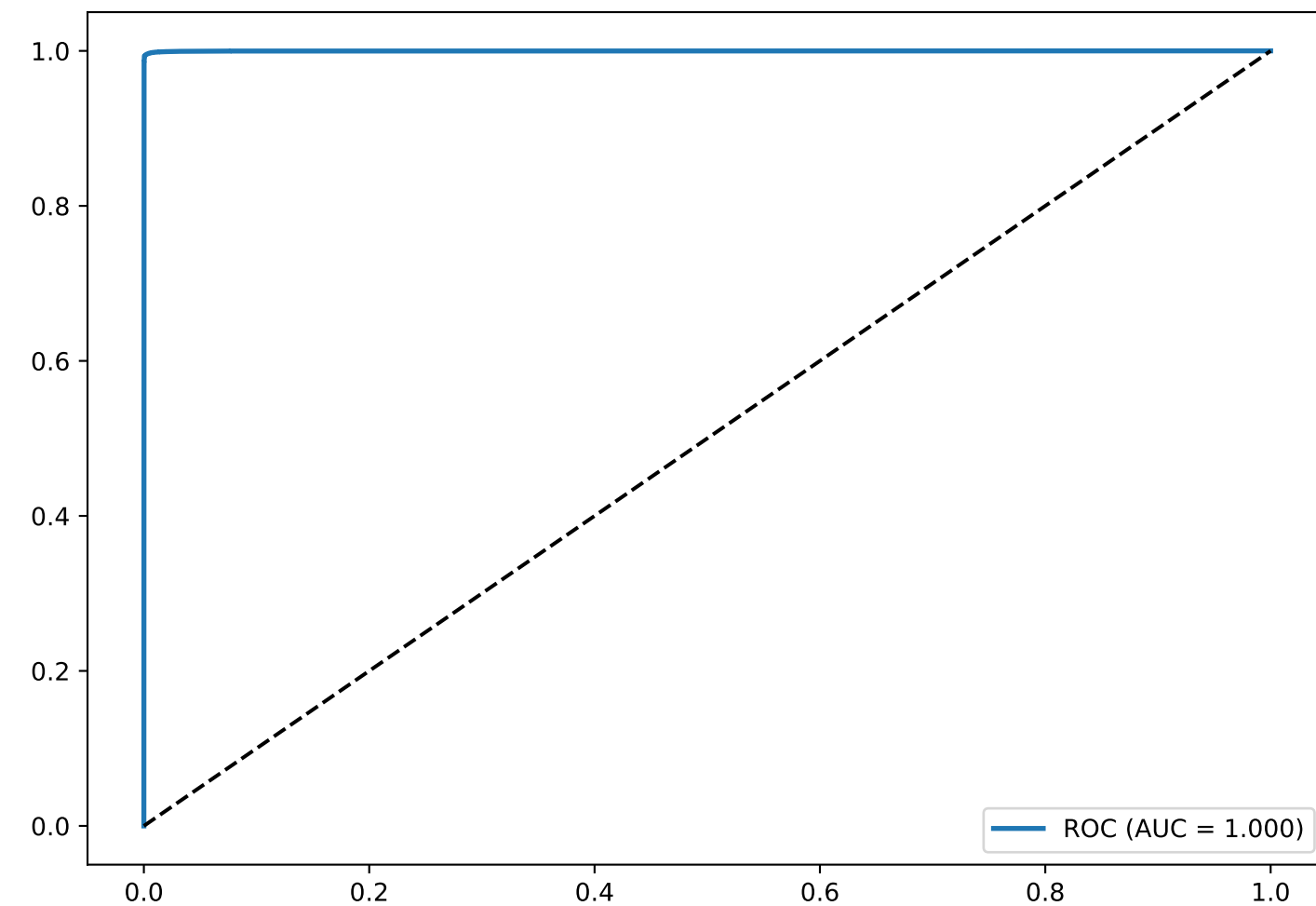


# EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.997

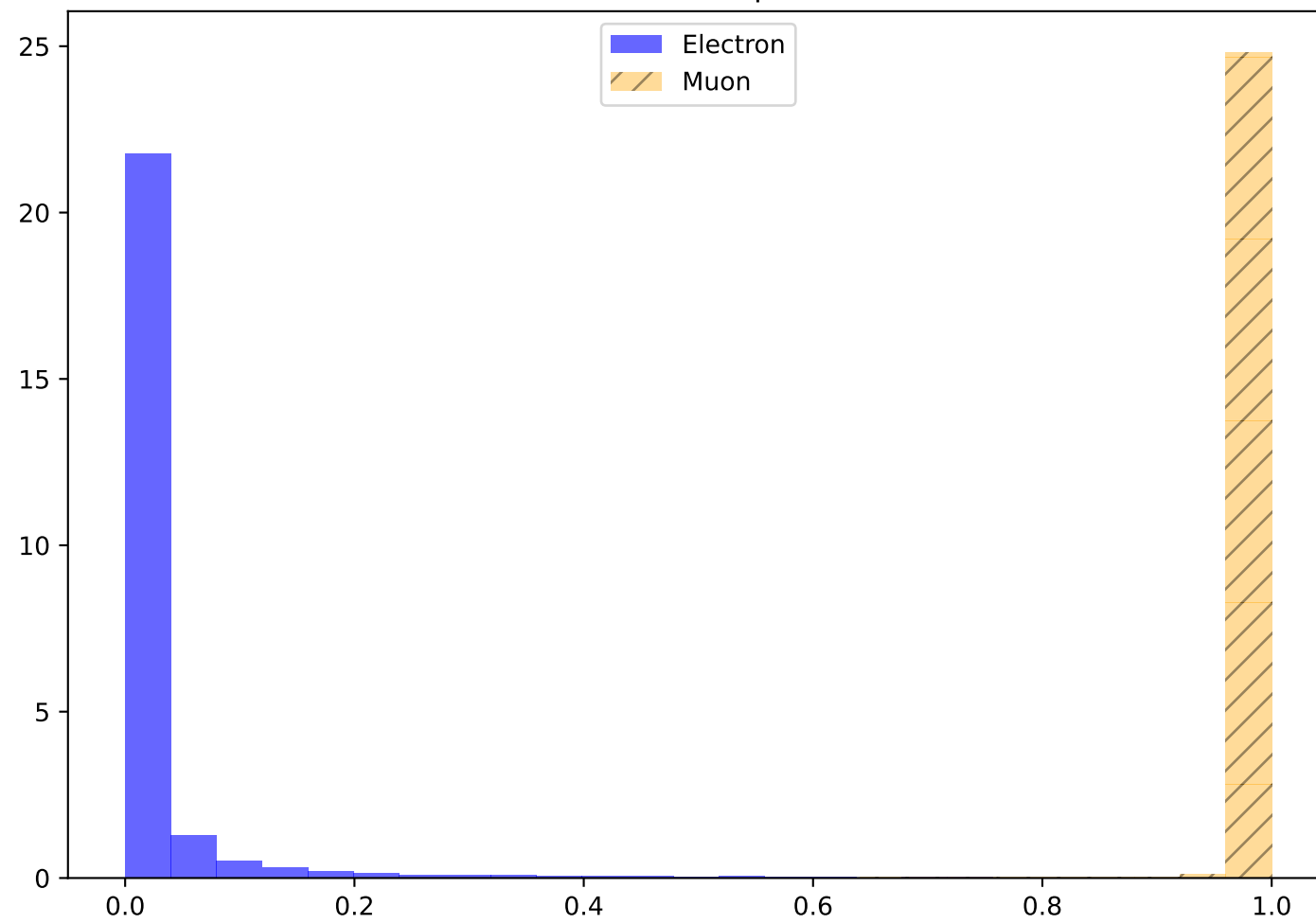
Confusion Matrix



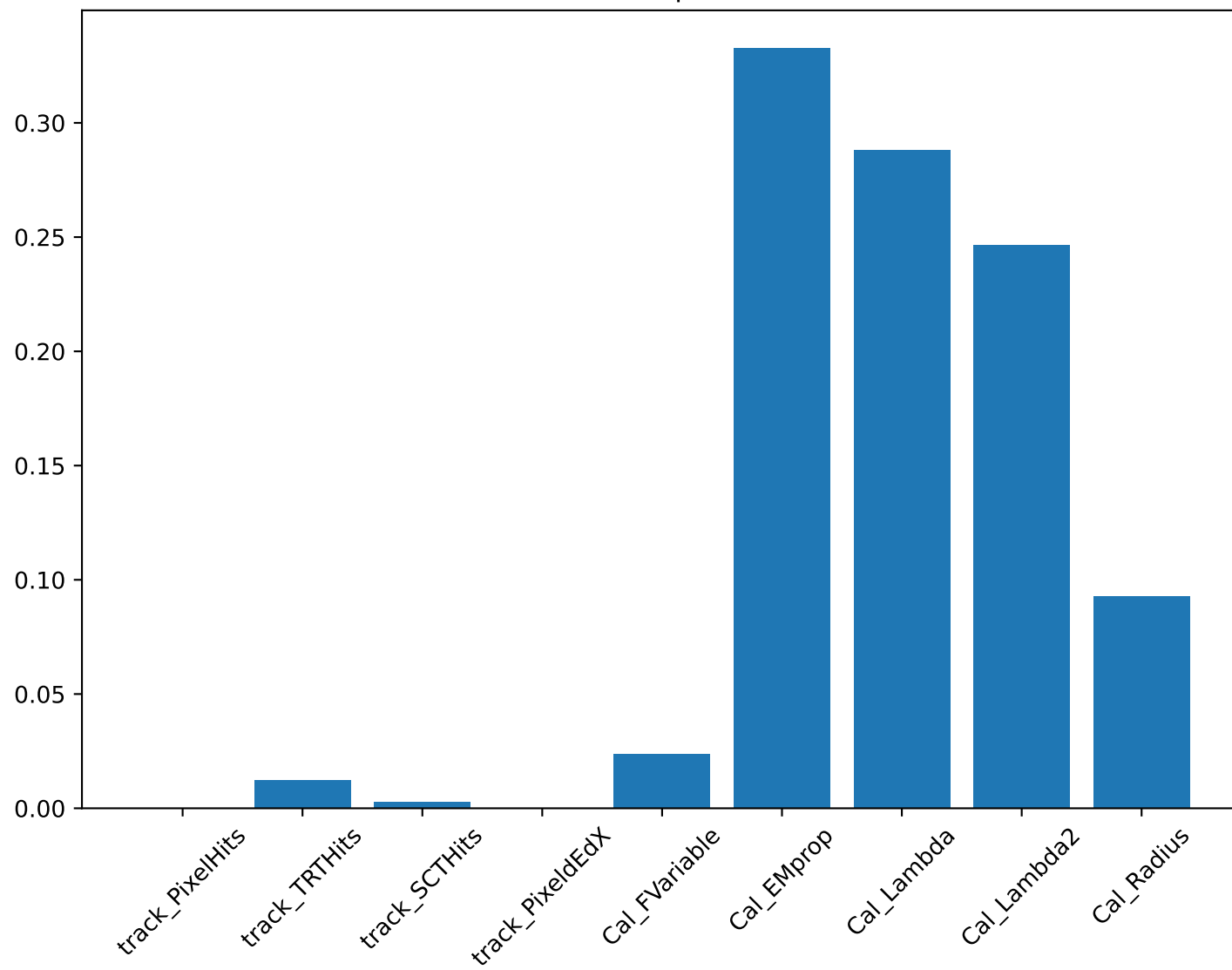
ROC Curve



Classifier Response

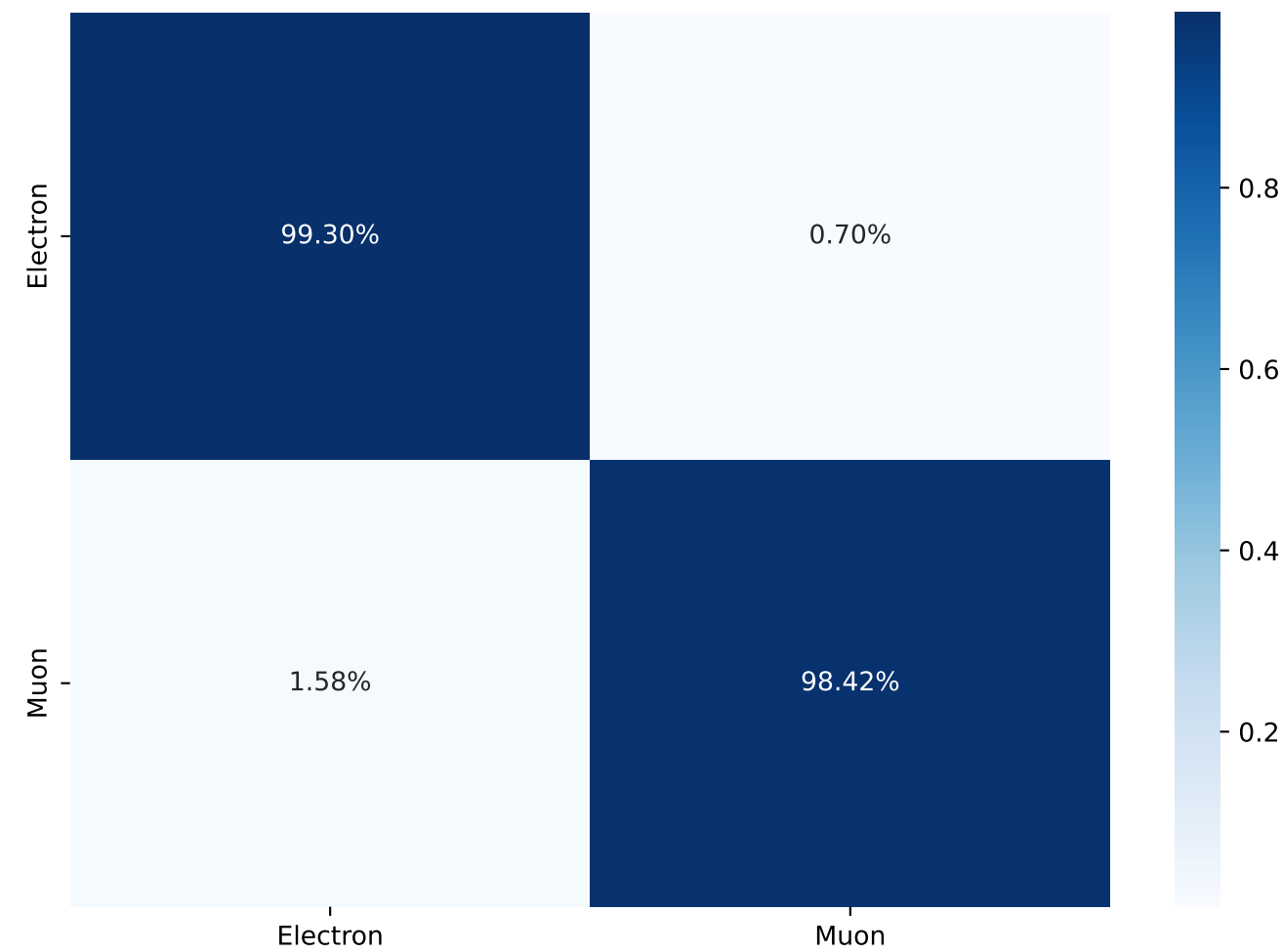


Feature Importance

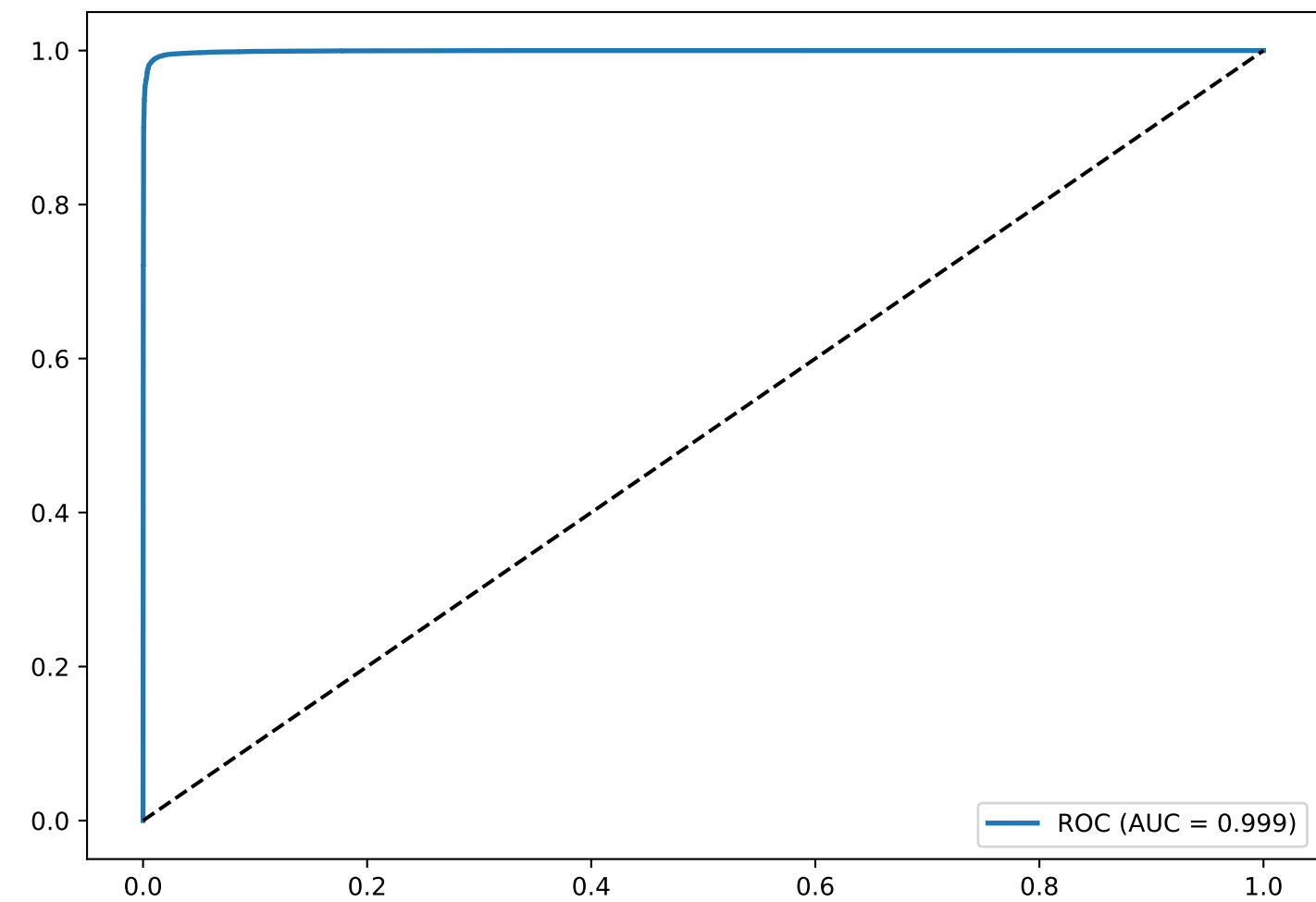


# EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.986

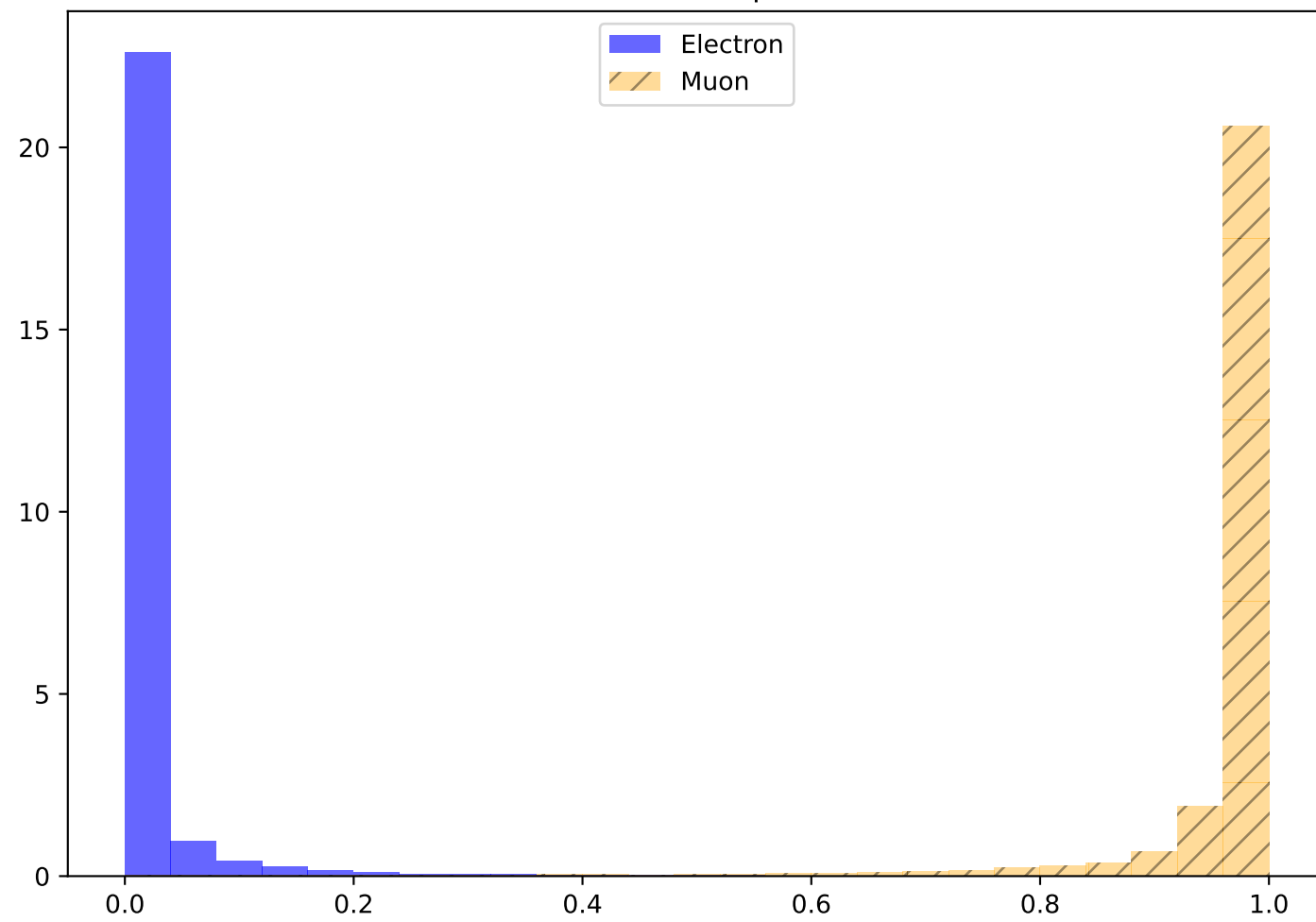
Confusion Matrix



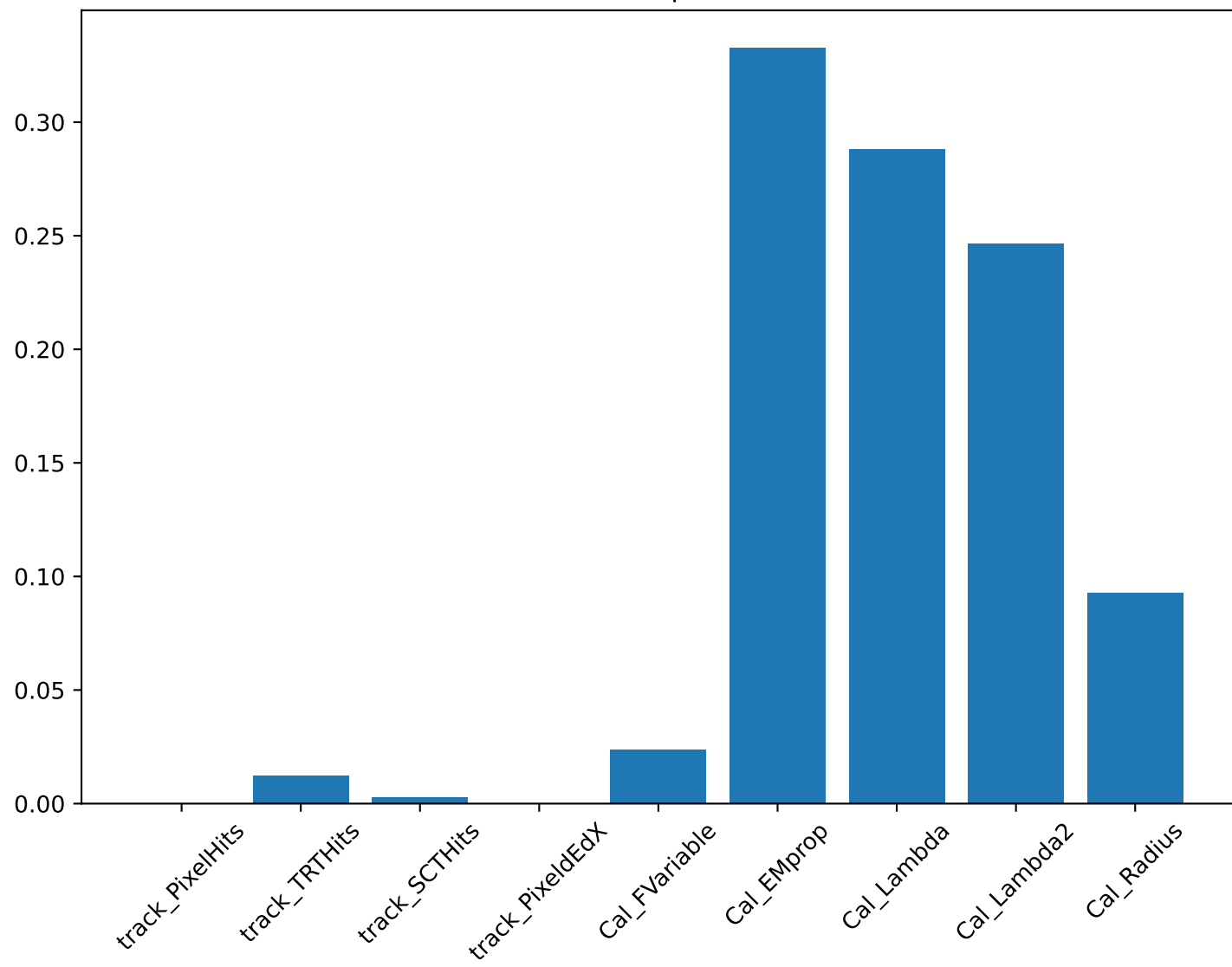
ROC Curve

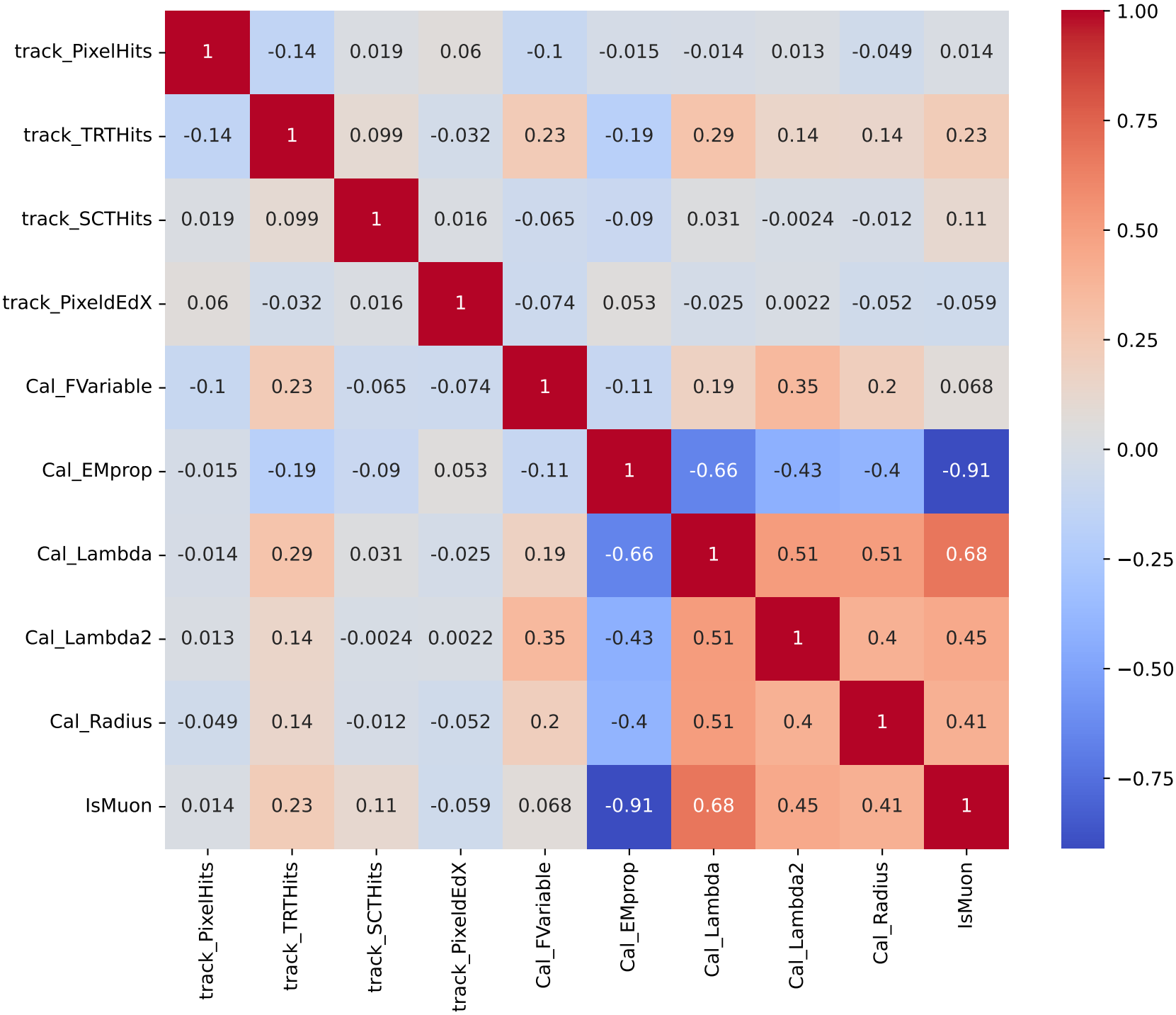


Classifier Response



Feature Importance





# Classification Reports Summary

## Trained: MCData.root | Test: MCData.root

### EtaRange = FullRange

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.98      | 0.99   | 0.99     | 89245   |
| Muon         | 1.00      | 0.99   | 1.00     | 278866  |
| accuracy     |           |        | 0.99     | 368111  |
| macro avg    | 0.99      | 0.99   | 0.99     | 368111  |
| weighted avg | 0.99      | 0.99   | 0.99     | 368111  |

### EtaRange = <1.0

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.98      | 1.00   | 0.99     | 60102   |
| Muon         | 1.00      | 0.99   | 1.00     | 158614  |
| accuracy     |           |        | 0.99     | 218716  |
| macro avg    | 0.99      | 0.99   | 0.99     | 218716  |
| weighted avg | 0.99      | 0.99   | 0.99     | 218716  |

### EtaRange = 1.0-1.5

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.99      | 0.99   | 0.99     | 15779   |
| Muon         | 1.00      | 1.00   | 1.00     | 73776   |
| accuracy     |           |        | 1.00     | 89555   |
| macro avg    | 1.00      | 0.99   | 0.99     | 89555   |
| weighted avg | 1.00      | 1.00   | 1.00     | 89555   |

### EtaRange = 1.5-2.5

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Electron     | 0.95      | 0.99   | 0.97     | 13364   |
| Muon         | 1.00      | 0.98   | 0.99     | 46476   |
| accuracy     |           |        | 0.99     | 59840   |
| macro avg    | 0.97      | 0.99   | 0.98     | 59840   |
| weighted avg | 0.99      | 0.99   | 0.99     | 59840   |